COS221
L05 - Enhanced Entity-Relationship Modelling (Part 1)

(Chapter 8 in Edition 6 and Chapter 4 in Edition 7)

18 February 2026

Semantic data modelling in Computer Science

- ER works well for traditional business apps, but struggle with complex domains(CAD/CAM, telecoms, GIS)
- Various semantic data models have been introduced into computer science
- Cross-field origins
  - Knowledge representation in AI
  - Object modelling in SE
- **Enhanced ER(EER) for Database systems.**

  This includes concepts such as:
  - **Superclass/subclass** relationships
  - **type inheritance**
  - **specialisation/generalisation** and constraints
  - modelling **UNION types or categories**
  - …

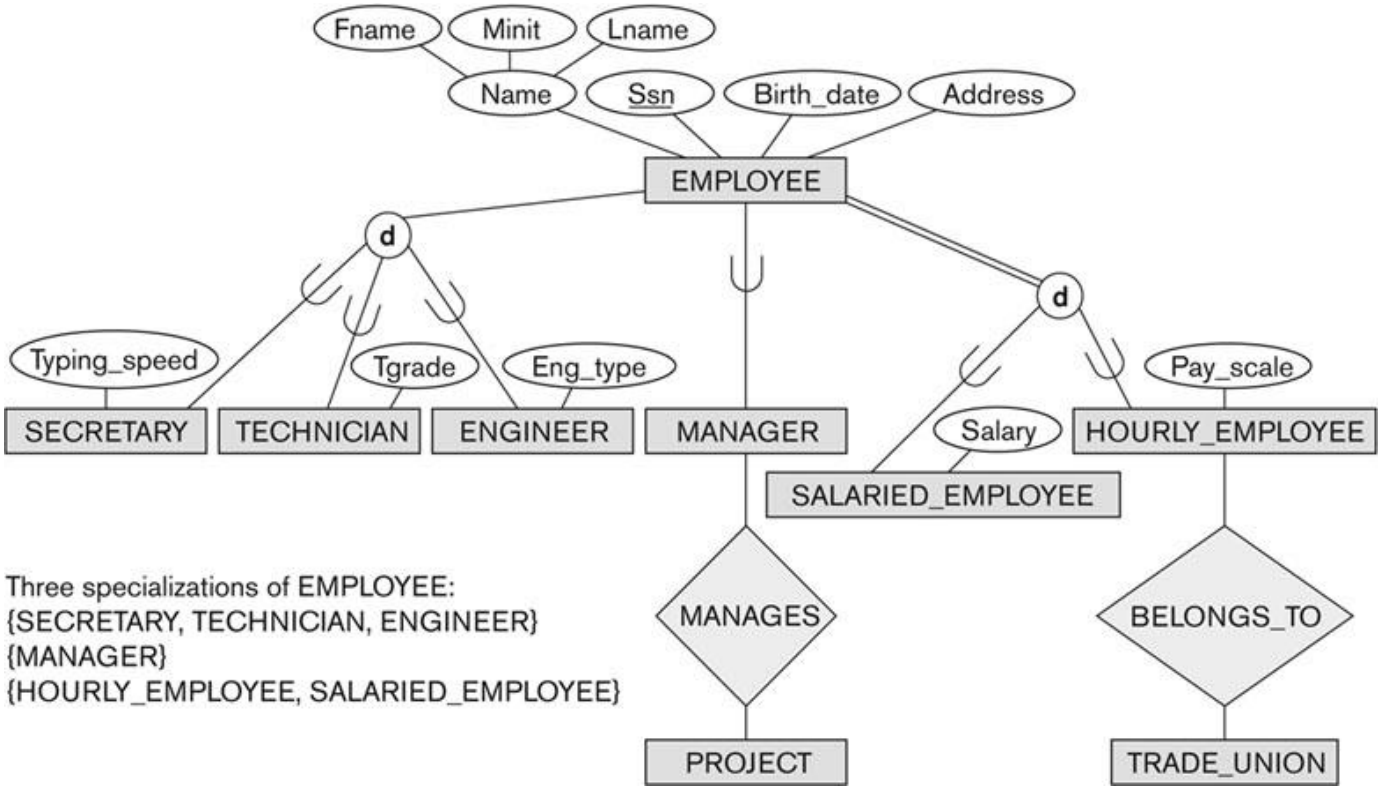
Subclasses and Superclasses

- An entity type may have additional meaningful subgroupings of its entities-> model as subclasses / subtypes
- **-Example**: EMPLOYEE may be further grouped into:
    - SECRETARY, ENGINEER, TECHNICIAN, …
    - MANAGER
    - SALARIED_EMPLOYEE, HOURLY_EMPLOYEE

- EER diagrams extend ER diagrams to represent these additional subgroupings, called **subclasses** or **subtypes**

Subclasses and Superclasses

- **Figure 4.1** EER diagram notation to represent subclasses and specialization.



Subclasses and Superclasses

- The set or collection of entities in each of the subgroupings is a subset of the entities that belong to the Employee entity set meaning every entity that is a member of these subgroupings is also an employee.
- Each of these subgroupings is a subset of EMPLOYEE entities
- We call each of these subgroupings a subclass or subtype of EMPLOYEE entity type
- EMPLOYEE is the superclass for each of these subclasses
- The relationship between superclass and any of its subclasses is called superclass/subclass (supertype/subtype or class/subclass relationship) relationships:

  - **EMPLOYEE/SECRETARY**
  - **EMPLOYEE/TECHNICIAN**
  - **EMPLOYEE/MANAGER**
  - …

Subclasses and Superclasses

- These subclass-superclass relationships are also called **I S-A** relationships
  - SECRETARY I S-A EMPLOYEE, TECHNICIAN I S-A EMPLOYEE, ….
- **Note**: An entity that is member of a subclass represents the same real-world entity as some member of the superclass:
  - The subclass member is the same entity in a **distinct specific role**
  - An entity cannot exist in the database merely by being a member of a subclass; it must also be a member of the superclass
  - It is not necessary that every entity in a superclass is a member of some subclass.

Subclasses and Superclasses

- **Examples:**
  - A salaried employee who is also an engineer belongs to the two subclasses:
    - **ENGINEER,** and
    - **SALARIED_EMPLOYEE**
  - A salaried employee who is also an engineering manager belongs to the three subclasses:
    - **MANAGER,**
    - **ENGINEER,** and
    - **SALARIED_EMPLOYEE**
- It is not necessary that every entity in a superclass be a member of some subclass

Attribute and Relationship Inheritance in Superclass / Subclass Relationships

- An entity that is member of a subclass inherits
  - All attributes of its superclass and all relationships in which in which the superclass participates **(type inheritence)**
- **Example:**
  - In the previous slide, **SECRETARY** (as well **as TECHNICIAN** and **ENGINEER**) **inherit** the attributes **Name, SSN, …,** from **EMPLOYEE**
  - Every **SECRETAR**Y entity will have values for the inherited attributes

Specialization

- **Specialization** is the process of defining a set of subclasses of a superclass
- The set of subclasses **is based upon some distinguishing characteristics** of the entities in the superclass
  - **Example**: {SECRETARY, ENGINEER, TECHNICIAN} is a specialization of EMPLOYEE based upon **job type**.
  - **Example:** MANAGER **is a specialization of EMPLOYEE based on the role the employee plays**
    - May have several specializations of the same superclass

Specialisation and Generalisation

A ***specialisation*** defines a set of subclasses of an entity type - **the superclass of the specialisation.**

The set of subclasses is **based upon some distinguishing characteristics of the entities in the superclass**

In the figure below there are three (3) specialisations of EMPLOYEE shown.

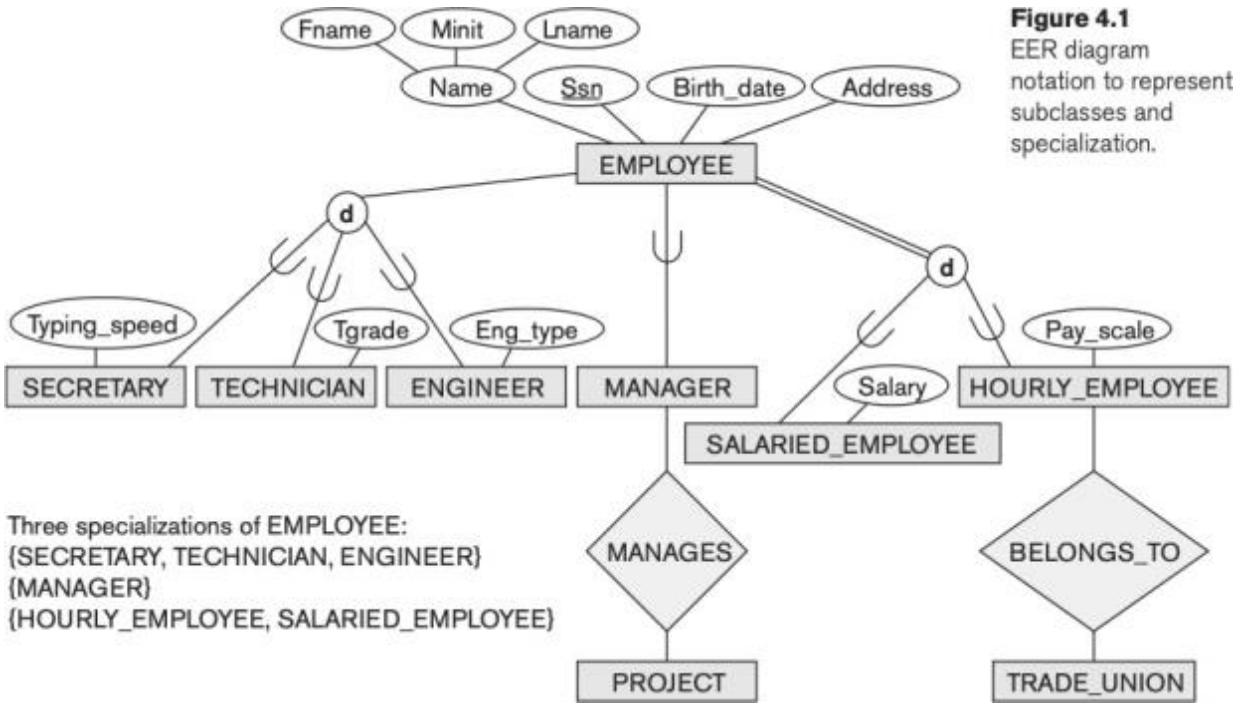


Figure 4.1
EER diagram notation to represent subclasses and specialization.

Specialisation and Generalisation - Notation

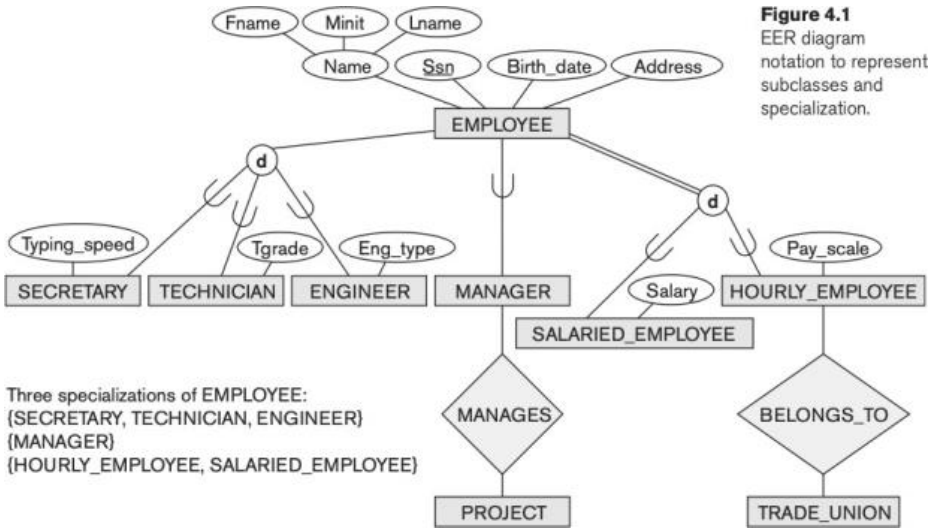


Figure 4.1
EER diagram notation to represent subclasses and specialization.

- Specialisations are linked to the superclass with a line to a circle then another line to the superclass.
- The ***proper subset*** *symbol* on the lines shows the direction of the subset/superclass relationship.
- Subsets may include attributes specific to the subclass, referred to as ***specific*** or ***local*** **attributes**.

Specialisation and Generalisation - Notation

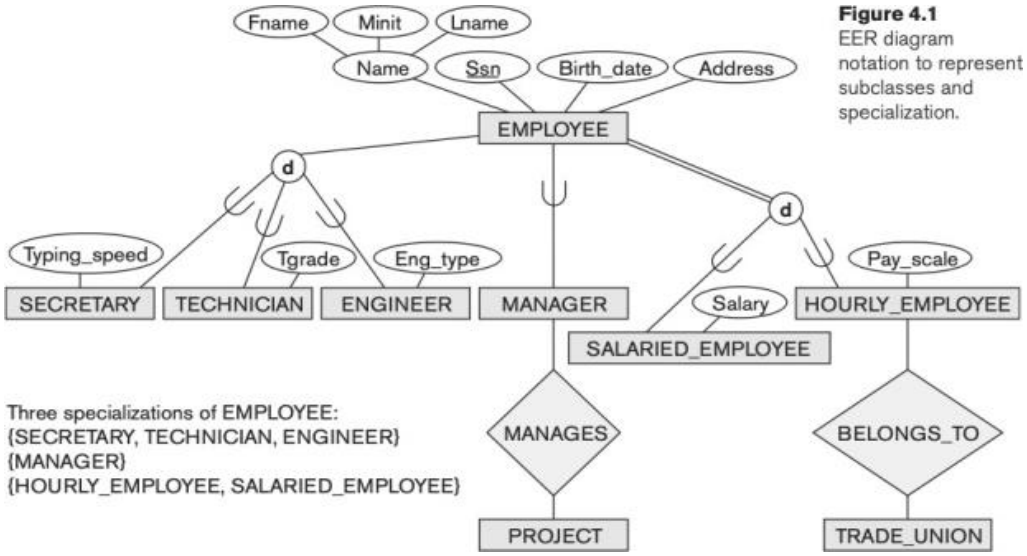


Figure 4.1
EER diagram notation to represent subclasses and specialization.

- A specialisation may comprise of multiple entity types with the same characteristics **(SECRETARY, TECHNICIAN** and **ENGINEER**) or a single entity type **(MANAGER)**
- A specialisation with a single entity type is drawn using only a line. That is, the circle notation is not used.

Specialisation and Generalisation - Notation

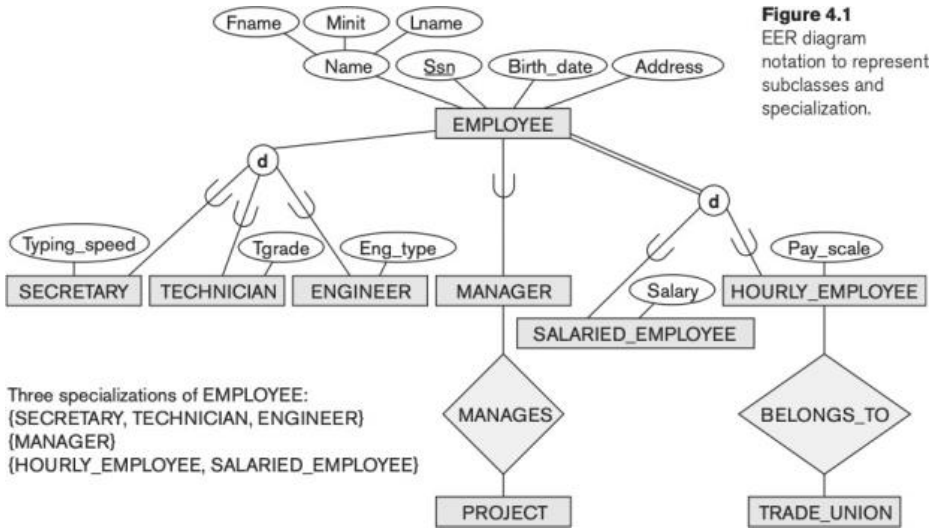


Figure 4.1
EER diagram notation to represent subclasses and specialization.

- Specialisations are used when some relationships may be participated in and others not.
- An **HOURLY_EMPLOYEE** may belong to a trade union.

Specialisation and Generalisation - Instances of a Specialisation

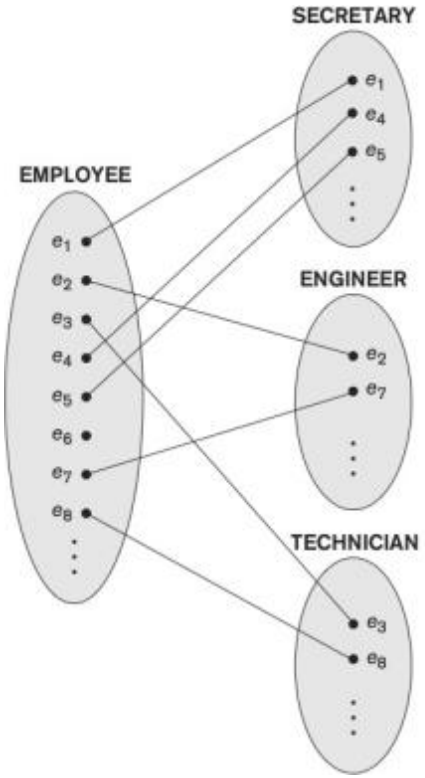


Figure 4.2
Instances of a specialization.

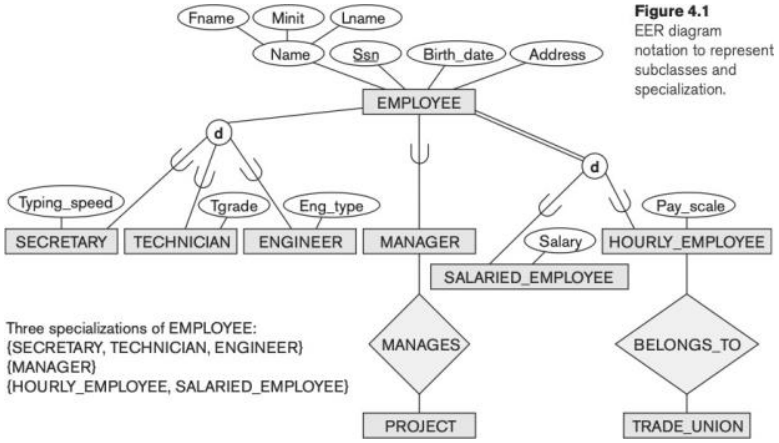
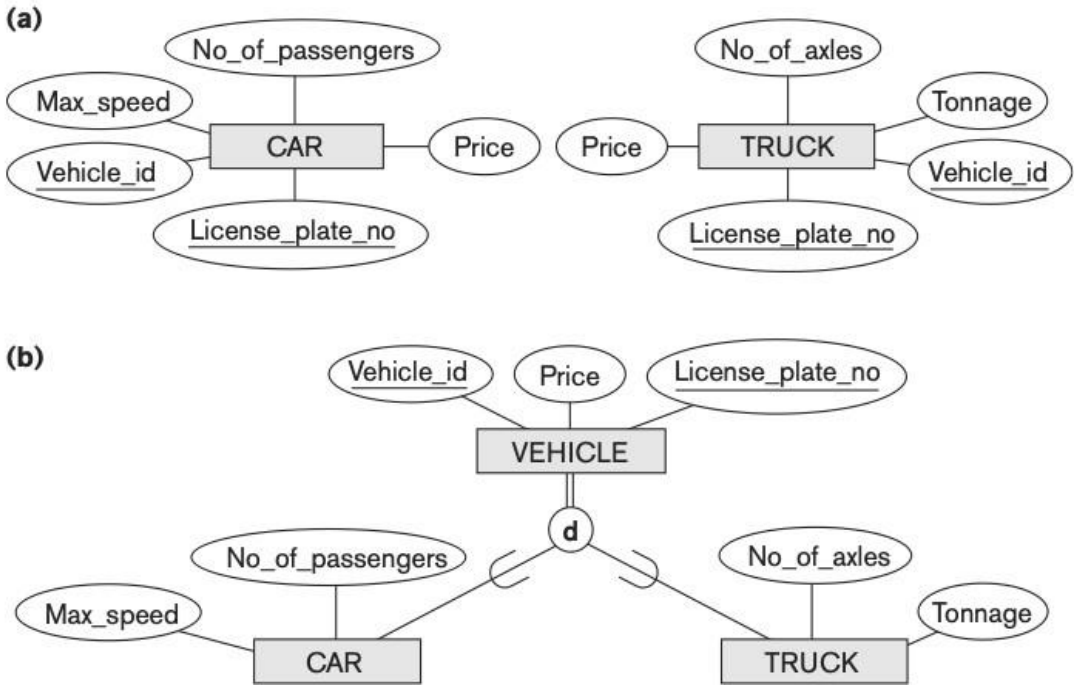


Figure 4.1
EER diagram notation to represent subclasses and specialization.

Specialisation and Generalisation

Figure 4.3
Generalization. (a) Two entity types, CAR and TRUCK.
(b) Generalizing CAR and TRUCK into the superclass VEHICLE.



- The reverse process of specialisation is **generalisation**.
- Given two or more related entities that share common attributes, these common attributes can be **generalised (abstracted out)** to form the superclass.

Constraints on Specialisation/Generalisation Hierarchies

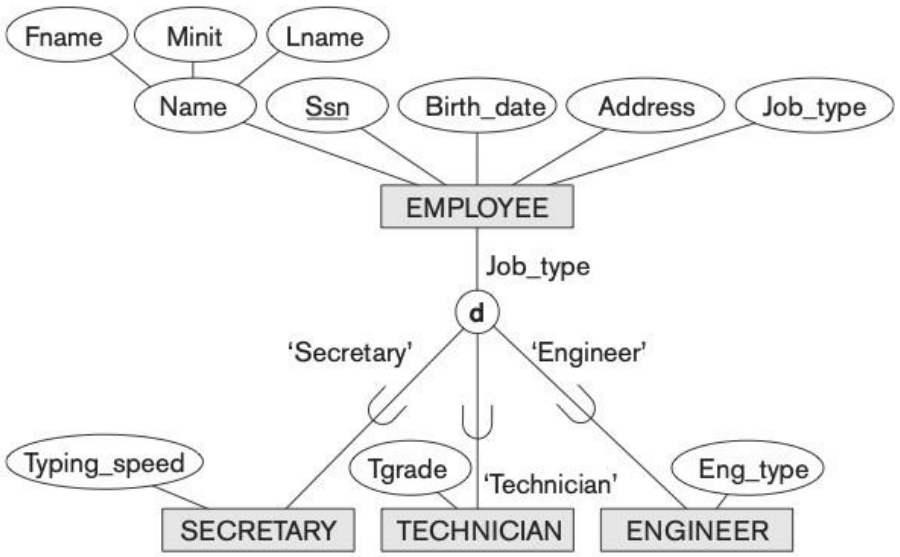


Figure 4.4
EER diagram notation for an attribute-defined specialization on Job_type.

- Placing a constraint on the value of some member of the superclass will allow us to specify exactly which entities will become members of the subclass.
- These subclasses are referred to as ***predicate-defined*** (or ***condition-defined*** subclasses.
- **`Job type = 'Secretary'`**, is called the ***defining predicate*** and **`Job_type`** in EMPLOYEE, the ***defining attribute***.
- The specialisation if termed an ***attribute-defined specialisation***.

Constraints on Specialisation/Generalisation Hierarchies

- When no condition is defined to determine membership of subclasses, the subclass is called ***user-defined.***
- Deciding membership of the subclass in this case is a manual process and determined by the user.
- The process cannot be automated.

Constraints on Specialisation/Generalisation Hierarchies

- Other constraints that may apply to a specialisation
  - **Disjointness** (or **disjointedness**)
  - **Completeness** (**or totalness**)
- **Disjointness** and **Completeness** are independent and therefore the following combinations are possible:
  - **Disjoint and Total**
  - **Disjoint and Partial**
  - **Overlapping and Total**
  - **Overlapping and Partial**

Constraints on Specialisation/Generalisation Hierarchies - Disjointness

- **Disjointness:** An entity of the specialisation may be a member of at most one subclass of the specialisation.
- An ***attribute-defined* specialisation implies disjointedness**.
- Disjointedness is specified using a **d** in the circle.
- Sets of entities of subclasses **that are not disjoint are said to be *overlapping***. An **o** in the circle is the designation for this, the default, case.

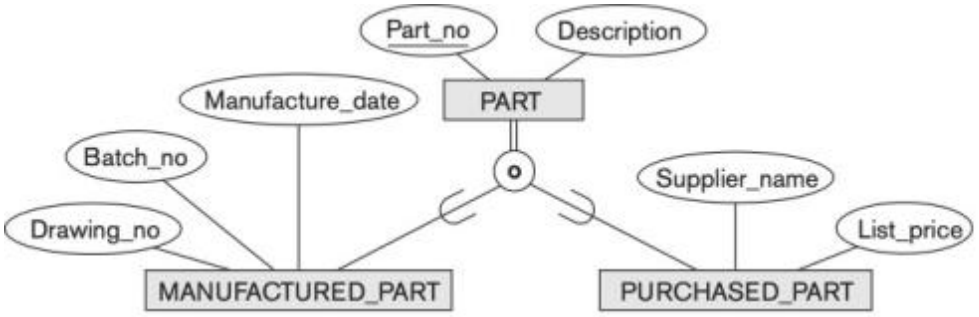
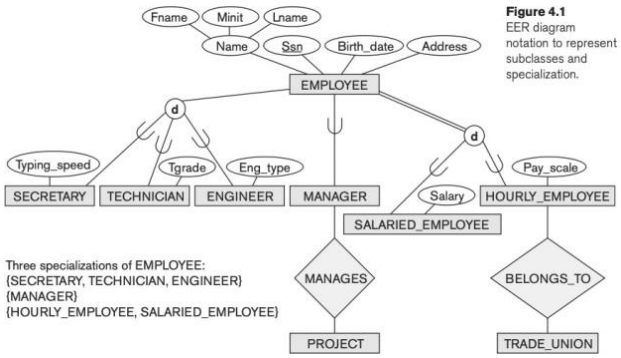


Figure 4.5
EER diagram notation for an overlapping (nondisjoint) specialization.

Constraints on Specialisation/Generalisation Hierarchies - Completeness

The completeness constraint may be *partial* or *total*.

- A *partial specialisation* means that the entity need not be a member of one of its subclasses.
  - This is shown on the ER-diagram using a single line.
  - For example, an EMPLOYEE may be a SECRETARY, ENGINEER, TECHNICIAN or MANAGER.
- A ***total specialisation*** means that every entity in the superclass **must** be a member of at least one subclass.
  - This is shown using a double line on the ER-diagram.
  - For example, an EMPLOYEE must be either a SALARIED_EMPLOYEE or an HOURLY_EMPLOYEE.



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EER diagram notation to represent subclasses and specialization.